

A Practical Approach To Verifying Structural Invariants In Colored Petri Nets

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- Existing symbolic techniques for SNs mainly apply to restricted subclasses.
- Tool support through SNexpression¹; a calculus for symbolic structural relations (conflict, causal connections, etc) is already available; recently lifted at the matrix level.

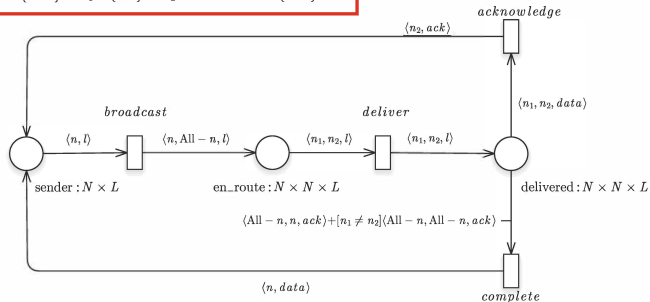
¹<http://www.di.unito.it/~depierro/SNexpression/>

Why (E)SN?

- SN: balance between expressivity and analytical capability due to a structured syntax.
- Behavioral symmetries can be exploited directly.
 - The symbolic reachability graph is strongly bisimilar to the ordinary one (lumped CTMC in Stochastic SN).
 - Symbolic discrete-event simulation
- **Extended SN**: provided with a calculus for symbolic structural analysis

Extended Symmetric Nets example: broadcasting protocol

`class L = {data} is L1 + {ack} is L2      class N = v{1..K}` K = 4



Key notions:

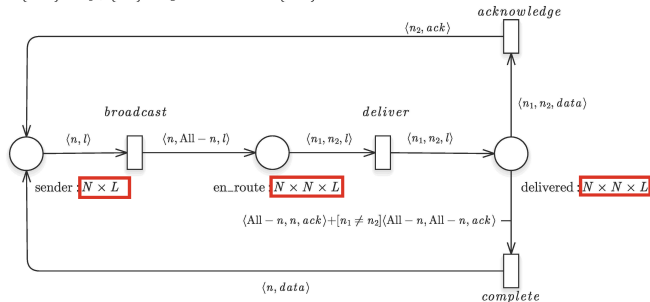
- color classes; can be partitioned or ordered circularly;
- color domains; products of color classes, defining token shapes in places and transition instances
- arc functions; sums of tuples of basic color functions (projections, successors, constants).

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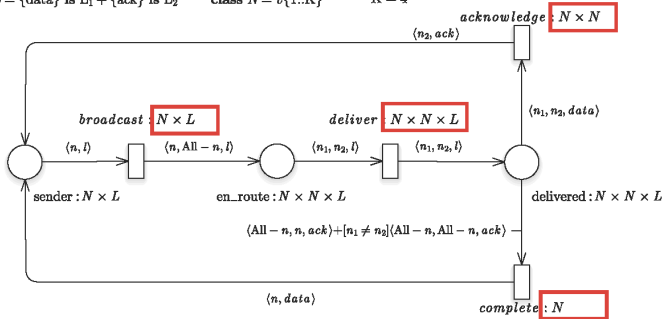


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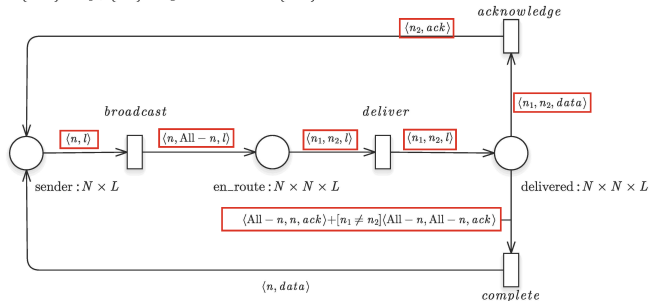
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m: P -vector such that $\mathbf{m}(p) \in \text{Bag}[\mathcal{D}(p)]$

A transition instance $c \in \mathcal{D}(t)$ is *enabled* in the marking **m** if:

$$\forall p \in P \ I(p, t)(c) \leq \mathbf{m}(p) \wedge (H(p, t)(c) = 0 \vee H(p, t)(c) > \mathbf{m}(p))$$

If c is *enabled* in **m**, it can fire, leading to $\mathbf{m}' := \mathbf{m} + O(p, t)(c) - I(p, t)(c)$

Arc-function expression

$$\langle All - n, n, ack \rangle + [n_1 \neq n_2] \langle All - n, All - n, ack \rangle \in \{N \rightarrow N^2 \times L\}$$

- maps each transition instance into a multiset of colored tokens.
- Interpretation: For a given $c \in N$, it returns all triples $\langle c', c, ack \rangle$ for every $c' \neq c$ (the first term), together with all triples $\langle c'', c''', ack \rangle$ for arbitrary c'', c''' such that $c'' \neq c$, $c''' \neq c$, and $c''' \neq c''$ (the second term).
- The guard that filters function evaluation makes ESNs more expressive than SNs.

The Language $\mathcal{L}$ of ESN arc functions

- Foundation of the symbolic structural calculus.
- Used to express conflict, causality and mutual exclusion.
- Implemented in SNexpression.

Role

Verification is reduced to symbolic manipulation inside $\mathcal{L}$.

A key enabler for symbolic manipulation.

Property

$\mathcal{L}$ is closed under all operators required by structural analysis.

$$\circ, \quad ^t, \quad +(-), \quad \ominus, \quad \cap$$

Flows Verification; Symbolic Incidence Matrix

Let $\mathbf{H}$ denote the $|P| * |T|$ symbolic incidence matrix of an ESN $\mathcal{N}$.

Each entry is a symbolic multiset expression² in $\mathcal{L}$.

$$\mathbf{H}[p, t] = O(p, t) - I(p, t) \in \{\mathcal{D}(t) \rightarrow Bag^*[\mathcal{D}(p)]\}.$$

The flow equations are verified symbolically, without unfolding.

² $Bag^*[\mathcal{D}]$: generalized bags, with multiplicities in $\mathbb{Z}$

P-semiflows and T-semiflows

P-semiflow $|P|$ -vector of functions in $\{\mathcal{D}(p) \rightarrow \text{Bag}[\mathcal{D}']\}$,
where $\mathcal{D}'$ is the shared place subdomain.

$$\mathbf{I}_P \cdot \mathbf{H} \equiv \mathbf{0}$$

T-semiflow $|T|$ -vector of functions in $\{\mathcal{D}' \rightarrow \text{Bag}[\mathcal{D}(t)]\}$,
where $\mathcal{D}'$ is the shared transition subdomain.

$$\mathbf{H} \cdot \mathbf{I}_T \equiv \mathbf{0}$$

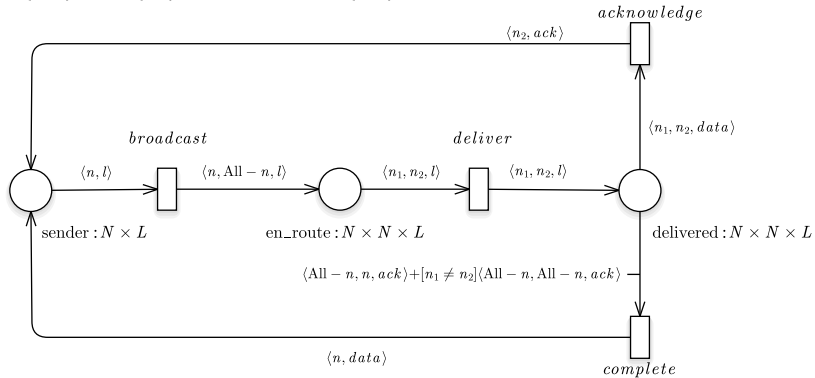
They characterize token conservation laws and "cyclic" transition sequences, capturing both qualitative and quantitative aspects. **General flows**: multisets with coefficients in $\mathbb{Z}$. **Duality**.

Running Example: T-Semiflow

class $L = \{\text{data}\}$ is $L_1 + \{\text{ack}\}$ is L_2

class $N = v\{1..K\}$

$K = 4$



Running Example: T-Semiflow

$$\mathbf{I}_T \left[\begin{array}{cc} \text{broadcast} & \text{deliver} & \text{acknowledge} & \text{complete} \\ (\langle n, data \rangle + & (\langle n, All - n, data \rangle + & \langle n, All - n \rangle_N & \langle n \rangle_N \\ \langle All - n, ack \rangle)_N & \langle All - n, n, ack \rangle + & & \\ [n_1 \neq n_2] \langle All - n, All - n, ack \rangle)_N & & & \end{array} \right]$$

Interpretation: It corresponds to the completion of a basic protocol cycle for any n representing the original sender.

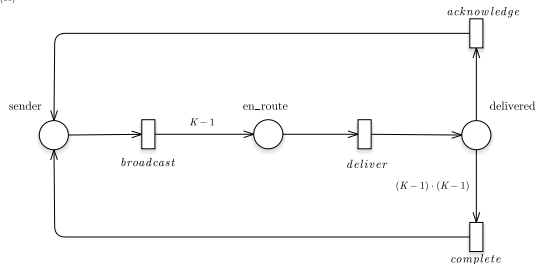
From Skeleton Flows to Symbolic Flows

Each ESN $\mathcal{N}$ can be equivalently rewritten so that all arc-functions have constant size.

We can automatically derive the skeleton of $\mathcal{N}$ by replacing functions with their sizes

Intuitive **property**: Each symbolic flow has to match a conventional (numerical) flow

$\langle K \rangle$



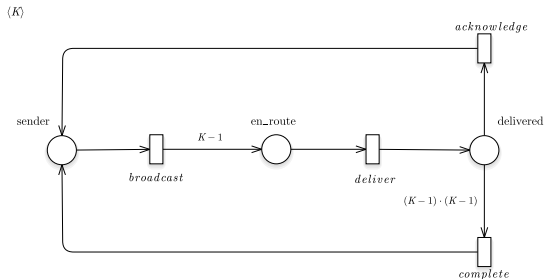
- Compute flows on the skeleton.
- Lift them to symbolic candidates.
- Verify them through calculus.
- It is *theoretically* possible to compute a flow basis, since arc functions of a given size can be algorithmically enumerated.

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Unique conventional T-semiflow (no P-semiflows); $K = |N|$

$[broadcast : K, deliver : K \cdot (K-1), acknowledge : K-1, complete : 1]$.

Beyond classical flows: Colored Semiflows

Flows presuppose fairly stringent structural requirements.

Colored flows abstract from multiplicities (only qualitative aspects); they are helpful for analyzing non-conservative or unbounded models.

They are based on $\overline{\mathcal{L}}$, the language of arc-function supports. Define

$$\mathbf{H}^+[p, t] = \overline{O(p, t) \ominus I(p, t)}$$

and

$$\mathbf{H}^-[p, t] = \overline{I(p, t) \ominus O(p, t)}.$$

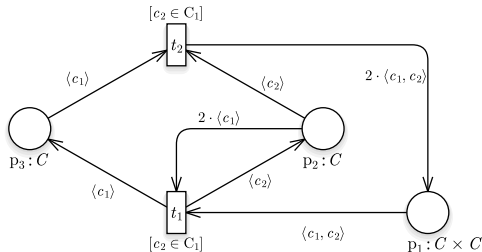
A colored semiflow $\mathbf{I}_{P-c}$ satisfies

$$\mathbf{I}_{P-c} \cdot \mathbf{H}^+ \equiv \mathbf{I}_{P-c} \cdot \mathbf{H}^-.$$

Colored Flow Verification: Example

class $C = v\{1..3\}$ is $C_1 + v\{4..N\}$ is C_2

$N = 10$



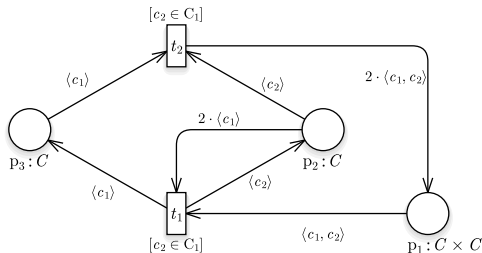
$$\checkmark \mathbf{I}_{p-c}^1 \quad \begin{array}{c} p_1 \\ [\langle c_2 \rangle [c_2 \in C_1]_{C,C} \end{array} \quad \begin{array}{c} p_2 \\ \langle c \rangle [c \in C_1]_C \end{array} \quad \begin{array}{c} p_3 \\ \end{array}]$$

$\mathbf{I}_{p-1}^1$: invariance of C_1 colors in place p_2 and in the second component of the tokens in p_1 .

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✓ I_{p-c}^2 p1 p2 p3
 $[\langle c_1 \rangle_{C,C}]$ $\langle c \rangle_C$

I_{p-c}^2 : the colors in place p_3 and in the first component of the tokens in p_1 remain invariant.

Verifying More General Invariants

Key Idea (see the example in the paper using inhibitor arcs)

Go beyond flows by expressing generic invariant properties as symbolic relations in $\mathcal{L}$.

- The invariant is represented using symbolic expressions, along with symbols denoting parametric markings and symbolic transition instances.
- Assuming the property holds initially, the verification proceeds inductively by checking that every *enabled symbolic transition* instance preserves it.
- Verification is performed through symbolic rewriting and inference within the calculus.
- Additional inference rules may be generated dynamically whenever required to complete the proof.

Flows become a special case of a more general inductive symbolic verification framework.

- ESN provides a flexible modeling formalism.
- $\mathcal{L}$ supports symbolic manipulation through closure properties.
- Symbolic and colored semiflows can be verified automatically.
- Skeleton flows guide candidate generation (and, from a perspective, basis generation).
- The framework naturally extends to more general invariants.
- Ongoing work (SNexpression):
 - full integration of multiset operators;
 - new functionalities for the verification of general structural invariants

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